



**TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
PURWANCHAL CAMPUS**

**A MINOR PROJECT REPORT ON
NEPALI LIFESTYLE BASED DIET AND FITNESS ADVISORY
SYSTEM**

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DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING
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**SUBMISSION DATE:
AUGUST 2026**

RECOMMENDATION

The undersigned certify that they have read and recommended to the Department of Electronics and Computer Engineering for acceptance, a project entitled “**NEPALI LIFESTYLE BASED DIET AND FITNESS ADVISORY SYSTEM**”, submitted by Himal Bhandari, Kshitiz Portel, Manav Acharya, and Mausam Parajuli in partial fulfillment of the requirement for the award of the degree of “**Bachelor of Engineering in Computer Engineering**”.

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August, 2026

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ABSTRACT

Diet and lifestyle-related diseases such as obesity, diabetes, and cardiovascular disorders are on the rise globally, and Nepal is no exception. Due to rapid urbanization, consumption of processed foods has become more common, and traditional eating habits are gradually being displaced, leading to various health disorders and nutritional imbalances. Most existing diet and fitness applications are based on Western food data and lifestyles, making them less suitable for Nepali users. The Nepali people's diet and daily routines are shaped by local ingredients, regional eating patterns, and cultural practices like religious fasting. This gap motivates the need for a health advisory system built specifically around the Nepali context. This project presents a web-based platform that delivers personalized dietary and fitness recommendations tailored to individual users. First, users set up their health profile, capturing details such as age, weight, medical conditions, and dietary constraints. Users can then log meals simply by taking a photo. The image is processed using a YOLO-based detection model that identifies and segments each food item on the plate, and a semantic matching engine, built on vector embeddings and cosine similarity, matches local food names like "bhat" to their standardized nutritional database entries. The system calculates the nutritional content of meals and checks it against the user's health profile to identify potential dietary concerns. This information is then provided to an LLM through a structured template to generate clear and consistent feedback. By combining food recognition, semantic search, and health-based filtering with Nepali food data, the system helps users understand their food choices and receive relevant dietary guidance based on their health and fitness goals.

Keywords: Nepali Food Recognition, Food Detection and Segmentation, YOLO, Nutritional Analysis, Semantic Food Matching, Personalized Diet and Fitness Recommendation, Large Language Model (LLM)

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LIST OF ABBREVIATIONS

API	Application Programming Interface
Colab	Colaboratory
mAP	mean Average Precision
YOLO	You Only Look Once
WHO SEAR	World Health Organization Regional Office for South-East Asia
R-CNN	Region-based Convolutional Neural Network
LLM	Large Language Model
RAG	Retrieval-Augmented Generation

CHAPTER 1

INTRODUCTION

1.1 Background

A healthy lifestyle depends on proper diet and regular physical activity. However, dietary and fitness needs are different for each person and depend on factors such as age, weight, activity level, health conditions, and food preferences. Many existing diet and fitness applications provide general recommendations and do not fully consider these individual needs.

In Nepal, food habits are strongly influenced by local culture and traditions. Common Nepali foods, dietary preferences, and practices such as fasting are often not properly considered in existing health applications. This makes it difficult for Nepali users to receive recommendations that match their daily lifestyle.

With the development of Artificial Intelligence and computer vision, it is possible to develop systems that can recognize food from images and provide personalized recommendations. The proposed Nepali Lifestyle-Based Diet and Fitness Advisory System aims to provide diet and fitness guidance based on the user's personal information and Nepali food habits. The system will also use computer vision to identify food items from meal images and use nutritional information to support better dietary recommendations.

1.2 Gap Identification

Many existing diet and fitness applications provide general health and nutrition recommendations, but they are mainly designed around international food items and eating patterns. They often have limited support for Nepali foods, local eating habits, and cultural practices such as fasting. Another limitation is that these systems may not consider important user-specific factors such as age, weight, activity level, health conditions, and dietary preferences when providing recommendations. Existing research also shows that many AI-based nutrition systems do not adequately address the cultural and dietary practices found in Nepal.

There is therefore a need for a system that combines Nepali food recognition, nutritional information, and personalized diet and fitness recommendations in one platform. The proposed system aims to address this gap by using a custom Nepali food dataset and AI-based food recognition along with user information to provide more relevant recommendations for Nepali users.

1.3 Motivation

Maintaining a healthy diet and fitness routine requires people to understand their food intake and make choices according to their individual needs. However, tracking food, finding its nutritional information, and deciding what to eat can be difficult without proper guidance. This becomes more challenging when the available food and nutrition information does not adequately reflect common Nepali foods and eating habits.

This motivated us to develop a system that can make food tracking and diet planning easier for Nepali users by working with Nepali food database, training YOLO model to recognize Nepali foods and generate its nutritional information and give suitable diet and fitness recommendations based on user's personal information.

The project therefore aims to combine Artificial Intelligence, computer vision, nutritional information, and user-provided data in a single system to provide practical and personalized lifestyle guidance for Nepali users.

1.4 Objectives

The primary objectives of this project are:

- To develop a system that detects and identifies food items from meal images and provides their nutritional information.
- To provide personalized diet and fitness recommendations based on the user's health profile, dietary intake, and lifestyle.

CHAPTER 2

LITERATURE REVIEW

This chapter presents an overview of previous research and developments in automated dietary tracking, computer vision-based food recognition, and AI-driven personalized nutrition. Various researchers have found how the lack of proper nutrition and its effects on people's health can be prevented using AI-driven personalized nutrition systems. Some services are already available for this, though in limited scope which propose different ways of making it work.

Dietary monitoring plays an important role in preventive health, managing chronic diseases, and personal fitness. Traditional tracking relies heavily on manual logging which can be tedious but recent advances in computer vision and artificial intelligence enable automated dish recognition and instant nutritional estimation. Coupled with personalized nutrition frameworks, these technologies allow software to tailor recommendations based on individual metabolic demands and health goals.

Despite these advancements, existing platforms face distinct limitations. Many rely on standardized Western food databases that fail to capture regional culinary diversity, such as local Nepali dishes. Others depend heavily on manual data entry, human consultation, or computationally expensive algorithms that are difficult to deploy on mobile devices. This chapter explores current studies addressing these challenges and establishes the theoretical foundations of the proposed system.

2.1 Related Work

Nutritional assessment in local contexts highlights clear public health demands. Chaudhary et al. [1] examined the relationship between dietary habits, lifestyle, and bone mineral density (BMD) among older adults in Kathmandu, finding that nutrient-poor diets directly correlated with impaired bone health. Sharma et al. [2] evaluated sixty-one commercial complementary foods in Nepal against WHO SEAR Nutrient Profile Model benchmarks, finding that none met the recommended energy and nutrient thresholds. These findings emphasize the need for accessible tools that increase awareness around the nutritional composition of locally consumed foods.

To address health management through software, Shanmugavel et al. [3] built a mobile platform for gestational diabetes mellitus (GDM) patients at Dhulikhel Hospital. While the application improved patient outcomes, its focus remained strictly clinical, relying on manual data entry and routine clinician involvement. This reliance on human expertise limits its scalability for general diet tracking, pointing to a need for more automated systems.

Tsolakidis et al. [4] surveyed AI applications in personalized nutrition, noting a shift from basic calorie counting toward systems that dynamically adapt to personal health targets. However, they observed that reliance on generic food databases limits effectiveness across regional diets. Mortazavi et al. [5] highlighted long-term engagement because of "the law of attrition," due to which users regularly abandon health platforms when logging becomes tedious. Raj [6] further identified data privacy, algorithmic bias, and system transparency as critical barriers to adoption. These insights indicate that an effective tool must balance accurate personalization with effortless user interaction.

Bhatta [7] reviewed machine learning strategies in diabetes management, focusing on predictive analytics and user-centric meal planning while acknowledging ongoing data quality issues. Salinari et al. [8] demonstrated that dynamic algorithms can effectively tailor interventions based on individual metabolic responses. However, hardware requirements remain a friction point. Jiang et al. [9] introduced DietGlance, a system using smart glasses with knowledge bases for real-time dietary feedback. While innovative, dependence on specialized wearables creates socioeconomic barriers, underscoring the need for solutions built on everyday smartphones.

Manandhar et al. [10] developed SmartCal, a computer vision system targeting common Nepali dishes using Mask R-CNN. Their work validated the feasibility of automated detection for regional foods like lentils, rice, and spinach. However, instance segmentation models like Mask R-CNN demand substantial computational power, resulting in high latency on mobile edge devices.

From these reviewed works, it is clear that the need and demand for AI-driven nutritional tools is growing but a key gap remains that current systems don't work on local foods, rely on expensive devices like in the case of DietGlance and are computationally expensive due to the use of Mask R-CNN in SmartCal system. This project bridges that

gap by pairing lightweight object detection with semantic search and localized nutritional datasets.

2.2 Related Theory

2.2.1 Personalized Nutrition and Calorie Estimation

Personalized nutrition refers to providing dietary recommendations according to the individual characteristics, requirements, and lifestyle of a user. Unlike generalized dietary recommendations, personalized nutrition considers factors such as age, weight, height, physical activity, dietary preferences, and other user-provided information.

In the proposed system, user information is used to provide recommendations that are more relevant to the individual's lifestyle and dietary requirements. The system also considers the user's recorded food intake when generating dietary guidance. Food can be recorded either through image-based food recognition or through manual entry. Once a food item is recorded, its corresponding nutritional information can be obtained from the nutritional database.

The nutritional information may include values such as calories, protein, carbohydrates, fat, and other relevant nutrients. The retrieved information is used as part of the user's dietary record and can subsequently be considered when generating personalized recommendations.

2.2.2 Computer Vision and Object Detection

Computer vision enables computers to extract useful information from digital images. In the proposed system, computer vision is used to identify food items from images uploaded by users. This reduces the need for users to manually enter every food item and provides an automated method of recording food intake.

Object detection is a computer vision task that identifies both the class and location of objects within an image. Unlike image classification, which assigns a label to an entire image, object detection can identify multiple objects within the same image. A detection model generally produces a bounding box, a class label, and a confidence

score for each detected object.

The bounding box represents the spatial location of an object within the image, while the class label identifies the detected food item. The confidence score indicates the model's confidence that the detected region belongs to the predicted class.

During object detection, multiple overlapping predictions may sometimes be generated for the same object. Non-Maximum Suppression (NMS) is used to remove redundant detections by retaining the most confident prediction among highly overlapping bounding boxes. These processes allow the system to identify individual food items from a meal image before their nutritional information is retrieved.

2.2.3 YOLO Architecture

You Only Look Once (YOLO) is a deep learning-based object detection architecture designed to detect objects efficiently within an image. YOLO performs object localization and classification within a single detection framework, making it suitable for applications where relatively fast inference is required.

A typical YOLO-based architecture consists of three major components: the backbone, neck, and detection head. The backbone extracts visual features from the input image. These features represent different characteristics of the objects present in the image. The neck processes and combines features from different levels to improve detection of objects at different scales. The detection head uses the extracted features to predict the locations and classes of objects in the image.

For the proposed system, a YOLO-based model is trained using a dataset containing relevant Nepali food categories. The trained model receives a meal image as input and detects the food items present in the image. The detected food names are then passed to the subsequent nutritional information retrieval process.

2.2.4 Nutritional Knowledge Base and Semantic Retrieval

A nutritional knowledge base stores information associated with different food items. Each food entry may contain information such as the food name, serving information, calories, protein, carbohydrates, fats, and other nutritional attributes. In the proposed

system, the knowledge base is used to associate detected or manually entered food items with their nutritional information.

Food names may be represented in different forms because users can refer to the same food using different names or descriptions. Semantic retrieval allows the system to identify relevant information based on the meaning of the input rather than requiring an exact textual match.

This process uses embeddings to represent textual information as numerical vectors. An embedding model converts a text input into a vector representation that captures its semantic characteristics. The resulting vectors can then be compared to identify semantically similar entries.

Vector-based retrieval is useful for handling variations in food names and user queries. For example, a user may describe a food using a different name or phrase from the one stored in the nutritional database. Semantic retrieval can help identify the corresponding nutritional record without requiring an exact string match.

2.2.5 Retrieval-Augmented Generation

Large Language Models (LLMs) are artificial intelligence models capable of understanding and generating natural language. In the proposed system, an LLM can be used to interact with users and generate dietary guidance based on the information available in the system.

However, relying solely on the pretrained knowledge of an LLM can result in responses that contain inaccurate or unsupported nutritional information. Retrieval-Augmented Generation (RAG) addresses this limitation by retrieving relevant information from an external knowledge base and providing it to the LLM as additional context.

When a user submits a query, the query is converted into an embedding and used to retrieve relevant information from the nutritional knowledge base. The retrieved information is then provided to the LLM as contextual information. The LLM uses this context together with the user's information and query to generate a natural-language response.

This approach allows the system to combine structured nutritional information with the

language generation capabilities of an LLM. It also allows the nutritional knowledge base to be updated independently from the language model, reducing the need to retrain the model whenever nutritional information is modified.

CHAPTER 3

METHODOLOGY

This chapter describes the approach used to develop the proposed Nepali Lifestyle-Based Diet and Fitness Advisory System. The methodology covers the main stages of the system, starting from acquiring a food image and identifying the food items to obtaining their nutritional information and generating personalized diet and fitness recommendations. The chapter also presents the main processing pipelines and the software development and system design models used during the development of the project.

3.1 Image Acquisition

The first step in the proposed system is to obtain information about the food consumed by the user. The system provides two methods for recording food. The user can either upload an image of a meal or enter the food name manually. Image-based input is mainly used to allow the system to identify food items automatically, while manual entry provides an alternative when an image is not available or when the user wants to record a food item directly.

For image-based input, the user selects a meal image through the web interface and uploads it to the system. The uploaded image is then passed to the food detection component, where the trained object detection model processes the image and identifies the food items present in it.

☐	ID	Completed				Annotated by	image	
☐	1	Aug 18 2026, 14:44:28	1	0	0	MA		{}
☐	2	Aug 18 2026, 14:46:17	1	0	0	MA		{}
☐	3	Aug 18 2026, 14:48:43	1	0	0	MA		{}
☐	4	Aug 18 2026, 14:51:04	1	0	0	MA		{}
☐	5	Aug 18 2026, 14:52:55	1	0	0	MA		{}
☐	6	Aug 18 2026, 14:54:57	1	0	0	MA		{}
☐	7	Aug 18 2026, 14:56:55	1	0	0	MA		{}

Figure 3.1: Dataset for Food Detection

3.2 Food Detection

Food detection identifies the food items present in an uploaded meal image. A YOLO-based object detection model is trained on a dataset of commonly consumed Nepali foods, with images annotated by food class and location. During prediction, the uploaded image is passed to the trained model, which identifies the food items and provides their class, bounding box, and confidence score. The detected food names are then passed to the next stage to obtain their corresponding nutritional information.



Figure 3.2: Annotation of Food Images for Model Training

```

from ultralytics import YOLO # Load YOLO library

# Load lightweight YOLO11 Nano Segmentation model
model = YOLO("yolo11n-seg.pt") # Load pre-trained weights

# Fine-tuned training pipeline engineered for small dataset sizes (~300 images)
results = model.train( # Start training process
    data="/content/final_yolo_model_dataset/data.yaml", # Dataset config path
    task="segment", # Instance segmentation task
    epochs=120, # Max training cycles
    imgsz=640, # Image resolution
    batch=16, # Batch size
    workers=4, # CPU data loaders
    patience=25, # Early stopping limit

    # Optimization strategy
    optimizer="AdamW", # AdamW optimizer
    lr0=0.0015, # Initial learning rate
    lrf=0.01, # Final learning rate ratio
    cos_lr=True, # Cosine learning schedule
    weight_decay=0.0005, # Weight decay penalty

    # Strategic augmentation suite tailored for food plate segmentation
    augment=True, # Enable augmentations
    hsv_h=0.015, # Hue variation
    hsv_s=0.6, # Saturation variation
    hsv_v=0.4, # Brightness variation
    degrees=15.0, # Rotation angle
    translate=0.1, # Image translation shift
    scale=0.5, # Zoom scale factor
    shear=2.0, # Shear distortion angle
    perspective=0.0001, # Perspective shift angle
    flipud=0.0, # Vertical flip chance
    fliplr=0.5, # Horizontal flip chance
    mosaic=1.0, # 4-image mosaic mix
    mixup=0.1, # Image mixup probability
    copy_paste=0.1, # Mask copy-paste chance

    # Output tracking
    project="/content/runs/segment", # Save folder location
    name="nepali_plate_optimized_run", # Run folder name
    exist_ok=True, # Overwrite existing folder
    plots=True # Generate metric plots
)

```

Figure 3.3: Training YOLO Model for Food Detection

3.3 Nutritional Information and Calculation

After the food items have been identified, the system obtains their nutritional information from the nutritional database. The same process is used for food items entered manually by the user. The food name is matched with the appropriate record in the database so that information such as calories, protein, carbohydrates, fat, and other available nutrients can be obtained.

Since the same food may be referred to by different names, the system can use semantic matching to improve the retrieval process. This allows a food name provided by the

user or detected by the model to be associated with the relevant record even when the wording is not the same as the stored food name.

The retrieved nutritional information is then associated with the user's food record. When multiple food items are recorded, their nutritional information is considered together when analyzing the user's food intake.

3.4 Recommendation Generation

The final stage of the system is the generation of personalized diet and fitness recommendations. The recommendations are based on the user's personal information, recorded food intake, nutritional information, and other relevant lifestyle or dietary preferences provided to the system.

The retrieved information is used as context for the recommendation component. Where required, relevant information is retrieved from the nutritional knowledge base before being provided to the language model. This helps the system generate responses based on the information stored within the system rather than relying only on the general knowledge of the language model.

The generated response is presented to the user through the web interface and may include dietary suggestions, food-related guidance, and fitness recommendations according to the user's provided information.

3.5 System Block Diagram

The system block diagram illustrates the overall architecture of the proposed Nepali Lifestyle-Based Diet and Fitness Advisory System and shows how its major components interact with each other. The system is organized into different layers to separate user interaction, request handling, processing, data storage, and AI-based recommendation generation.

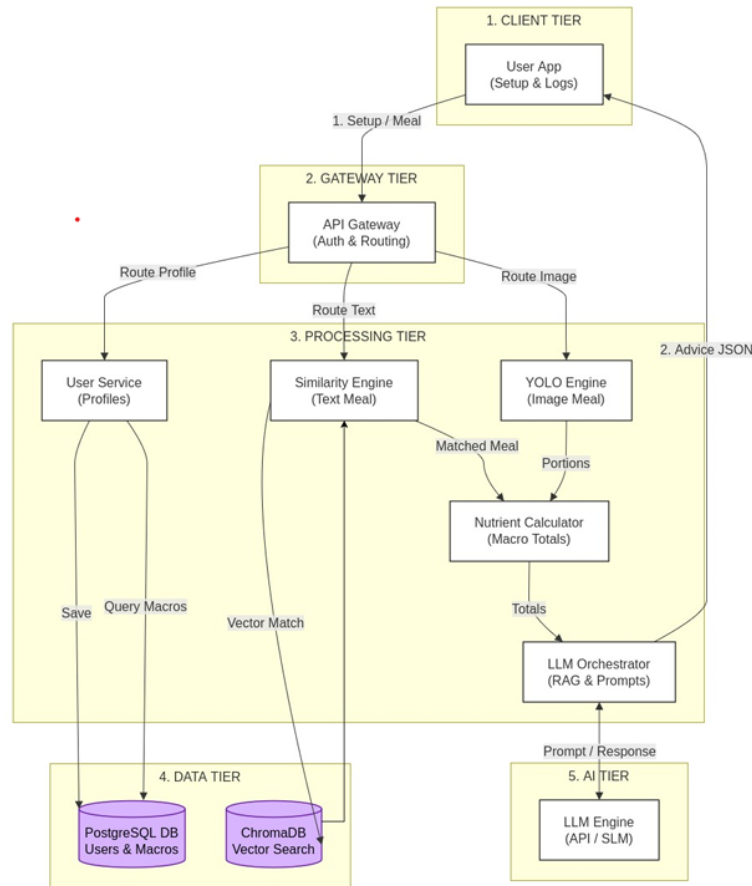


Figure 3.4: Block Diagram of System

The Client Tier provides the user interface through which users enter their profile information, health-related details, and meal information. Users can provide meal descriptions or upload images of food for analysis. The Gateway Tier acts as the entry point to the system and handles request routing and authentication before forwarding requests to the appropriate services.

The Processing Tier contains the main computational components of the system. The YOLO-based food detection model processes meal images and identifies the food items present. The nutrient calculator uses the detected food items and their portions to estimate the corresponding nutritional values. For text-based inputs, the similarity engine retrieves relevant food and nutritional information. The processed information is then passed to the LLM orchestrator, which combines the available data to generate personalized recommendations.

The Data Tier stores the information required by the system. PostgreSQL is used to

manage user profiles and related data, while ChromaDB is used for vector-based search and similarity matching. The AI Tier provides the language model used by the recommendation component to generate the final dietary and fitness advice.

Overall, the data flows from the user interface through the gateway and processing components, where food detection, nutritional calculation, and information retrieval are performed. The resulting information is then provided to the AI component to generate a personalized response, which is returned to the user through the client application.

3.6 Use Case Diagram

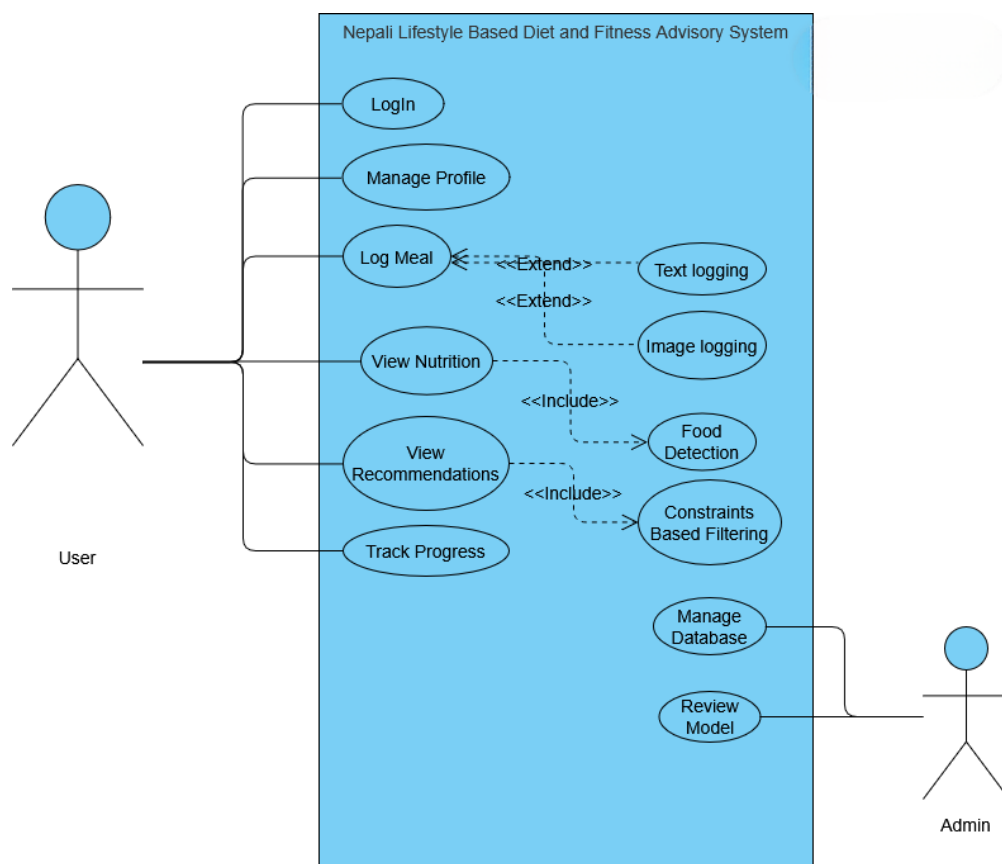


Figure 3.5: Use Case Diagram of the System

The Use Case Diagram illustrates the major functionalities of the system and the interactions between the two primary actors: the User and the Admin. The User can register and log in to the system, manage personal health information, log daily meals, view nutritional information, receive personalized diet and fitness recommendations, and monitor health progress. The Admin is responsible for managing the nutritional

database and monitoring the performance and accuracy of the system.

The Log Meal use case allows the user to record meals through either Text Logging or Image Logging. These two options extend the main meal-logging functionality, allowing the user to select the appropriate method for recording a meal. When the user accesses the View Nutrition functionality, the system performs Food Detection to identify food items from an uploaded meal image before retrieving the corresponding nutritional information.

Similarly, the View Recommendations use case incorporates Constraints-Based Filtering to evaluate the user's nutritional intake, personal profile, and applicable health conditions, such as diabetes or hypertension. Based on these factors, the system generates and displays personalized diet and fitness recommendations.

Overall, the use case design separates administrative functions from user-oriented activities while defining the major interactions required for meal logging, nutritional analysis, health monitoring, and personalized recommendation generation.

3.7 Sequence Diagram

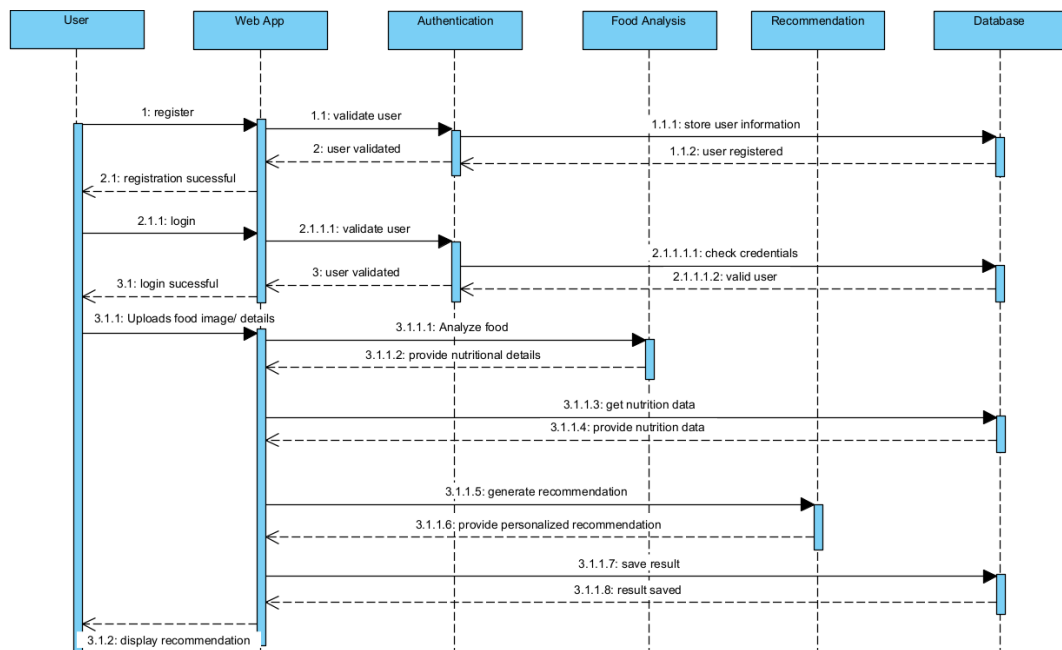


Figure 3.6: Sequence Diagram of the System

The sequence of operations is organized across six lifelines: User, Web App, Authentication, Food Analysis, Recommendation, and Database. The interaction follows three distinct operational phases.

At first, the User submits credentials through the Web App for validation through Authentication and storage in the Database. Secondly, the authenticated User uploads meal data, which the Web App routes to Food Analysis to identify food items, calculate nutritional metrics, and query baseline profiles from the Database.

Finally, the Web App passes the nutritional summary to the Recommendation module to synthesize a personalized diet plan, persist it in the Database, and display the final feedback to the user.

This design uses the Web App as a central mediator, enforcing transactional boundaries and ensuring that computationally intensive tasks are performed after session authentication.

3.8 Software Development Model

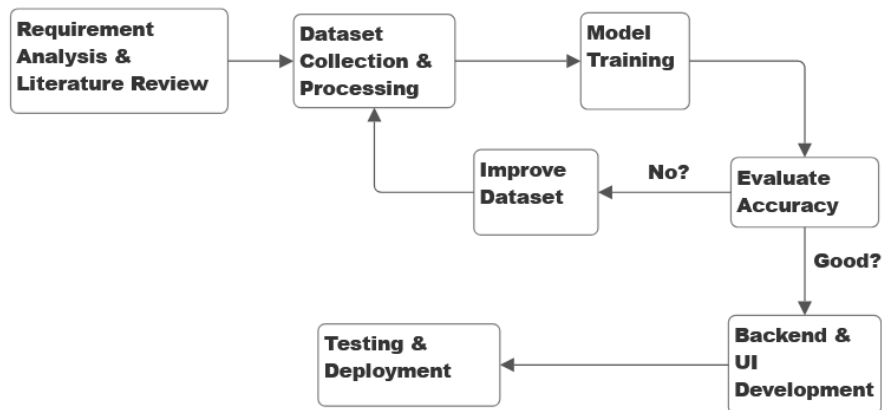


Figure 3.7: Software Development Model

The project follows a hybrid software development model combining the structured, sequential progression of the Waterfall model with the feedback-driven refinement of an iterative approach.

The workflow begins sequentially with Requirement Analysis & Literature Review, establishing core system objectives and functional specifications. It then transitions into Dataset Collection & Processing, followed by initial Model Training.

At the Evaluate Accuracy phase, a critical decision loop is introduced. If performance metrics are unsatisfactory, the process triggers an iterative cycle through Improve Dataset, feeding back into dataset collection and model retraining to optimize performance.

Once performance targets are met, the workflow proceeds sequentially to Backend & UI Development, integrating the validated machine learning pipeline with the user interface and system logic, before concluding with comprehensive Testing & Deployment.

CHAPTER 4

RESULT AND DISCUSSION

4.1 Results

The Nepali Lifestyle-Based Diet and Fitness Advisory System was developed as a web-based platform for providing personalized dietary and fitness guidance based on individual user information and local food practices. The system integrates user profile management, food detection and segmentation, nutritional information retrieval, health screening, and personalized recommendation generation.

The overall workflow begins with food detection and segmentation using the YOLO-based model. The detected food labels are then matched with the most relevant standardized nutritional records. The retrieved nutritional information is combined with the user's profile and dietary constraints to generate personalized dietary and fitness recommendations.

The developed system was evaluated through its major functional components, including user information management, food identification from meal images, nutritional analysis, and personalized recommendation generation. The results obtained from these components are presented in the following sections.

4.1.1 User Profile and Health Information

The system provides interfaces for user account registration and health profile setup. During registration, users provide their basic account information, including name, username, email address, and password. After creating an account, users complete their health and lifestyle profile by providing information such as age, gender, height, weight, target weight, activity level, fitness goal, and dietary preference. The collected information is used to support personalized dietary and fitness recommendations.

ACCOUNT SETUP

Create your account

Enter your details to register for personalized dietary guidance.

FIRST NAME *

LAST NAME *

Ram

Thapa

USERNAME *

ramthapa

EMAIL ADDRESS *

ramthapa12@gmail.com

PASSWORD *

CONFIRM PASSWORD *

Register

03 / PROFILE SETUP REQUIRED

Complete Your Profile

Before accessing your dashboard, please provide your body metrics and lifestyle preferences.

AGE *

30

GENDER *

Male

HEIGHT (CM) *

175

WEIGHT (KG) *

70

TARGET WEIGHT (KG)

75

Optional

ACTIVITY LEVEL *

Sedentary (little to no exercise)

FITNESS GOAL *

Maintain Weight

DIETARY PREFERENCE *

Vegetarian

Complete Profile Setup

Figure 4.1: User Account Registration and Health Profile Setup

4.1.2 Food Detection and Segmentation

The food detection and segmentation module identifies individual food items from a meal image using the trained YOLO-based instance segmentation model. The model processes the input image and predicts the corresponding food classes and segmentation masks, allowing each food item to be distinguished from the surrounding regions.

The model is trained using polygon-based annotations, which enable it to generate segmentation masks for the detected food items. The resulting segmented regions provide the basis for the subsequent processing and analysis of the identified food items.

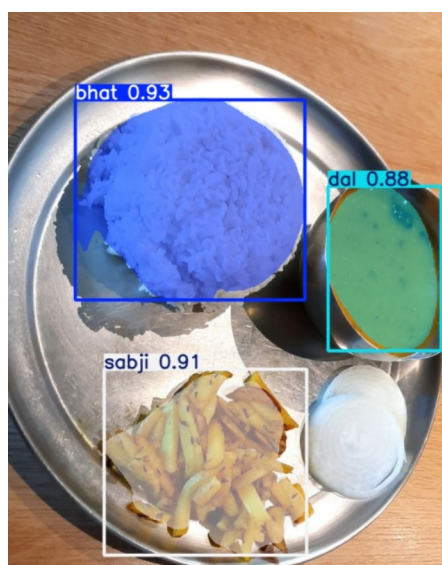


Figure 4.2: Food Detection and Segmentation Result

4.1.3 Nutritional Analysis

The identified food items were analyzed using the corresponding nutritional information retrieved from the database. The system presents the nutritional composition of the analyzed meal, including calories, carbohydrates, protein, fat, and other available nutritional values.

The segmented food regions obtained from the detection process are used to identify the respective food items and retrieve their associated nutritional information. The resulting nutritional data provides an overview of the meal's composition and serves as an input for the subsequent personalized diet and fitness recommendation process.

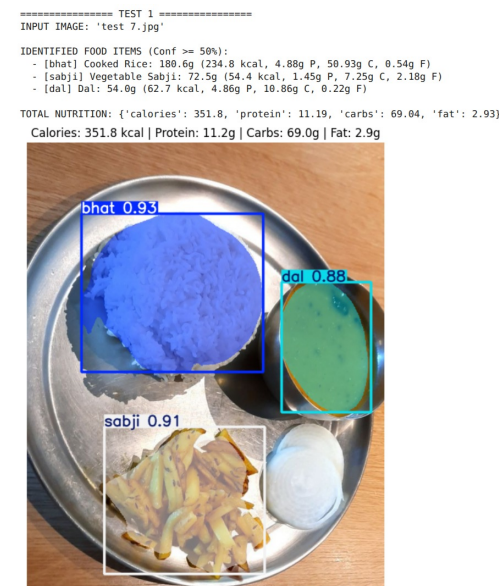


Figure 4.3: Nutrition Analysis Result

4.1.4 Personalized Diet and Fitness Recommendation

Based on the user's health and lifestyle information together with the analyzed nutritional data, the system generates personalized diet and fitness recommendations. The recommendations are tailored according to the user's fitness goals, dietary preferences, and available health information. The system provides suitable dietary guidance and fitness suggestions to support the user's individual needs and promote a healthier lifestyle.

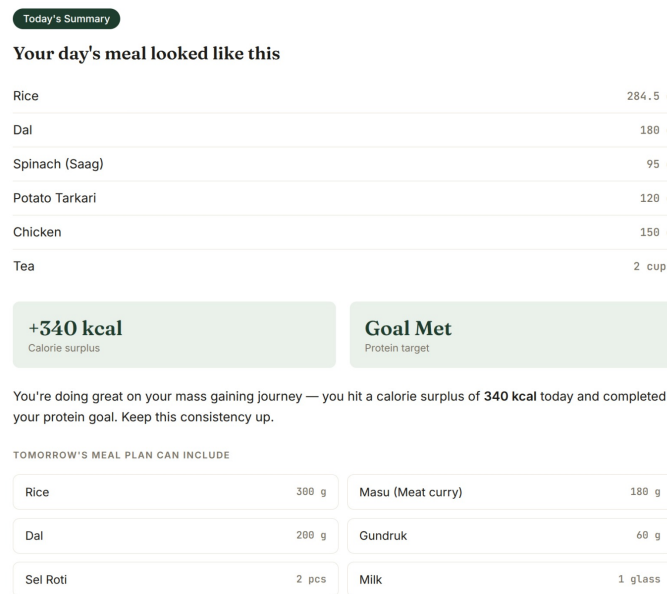


Figure 4.4: Personalized Diet and Fitness Recommendation Result

4.1.5 Health Report

The system generates a health report that summarizes the user's analyzed dietary and nutritional information. The report presents the identified food items, their corresponding nutritional values, and the personalized recommendations generated from the user's profile and meal analysis. It provides the user with a consolidated overview of the analyzed results, making the dietary information and recommendations easier to review.

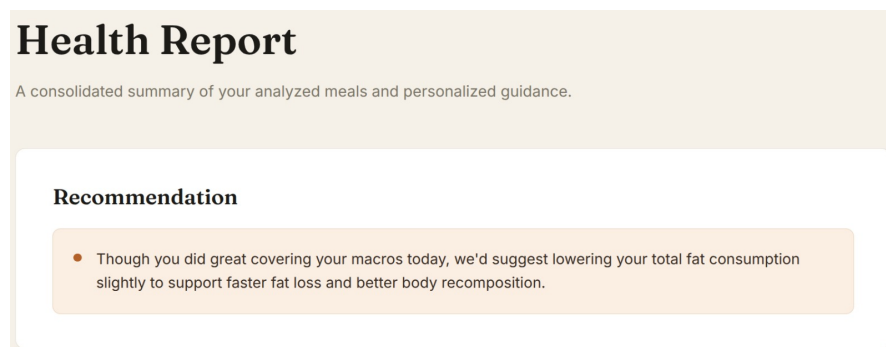


Figure 4.5: Health Report Generated by the System

4.2 Discussion

The results demonstrate the overall functioning of the developed Nepali Lifestyle-Based Diet and Fitness Advisory System across its major components. The system was able to process food images, identify and segment individual food items, retrieve the cor-

responding nutritional information, and use the analyzed data to generate personalized dietary and fitness recommendations. The integration of these components provides a complete workflow from food image analysis to personalized health and lifestyle guidance.

The food detection and segmentation component provides the foundation for the subsequent stages by identifying individual food items from meal images. The nutritional analysis then provides information about the composition of the identified food items, while the recommendation component uses the available user and nutritional information to provide personalized guidance. The generated health report presents the results in a structured form for the user.

Overall, the results indicate that the developed system can integrate food image analysis, nutritional analysis, and personalized recommendation into a single platform. However, the effectiveness of individual components depends on factors such as the quality and diversity of the food dataset, the accuracy of food identification, and the completeness of the nutritional information available in the database.

4.3 Model Training Results and Performance Analysis

The major components of the developed system were evaluated to assess their effectiveness and performance. The evaluation includes the training results of the YOLO-based food detection and segmentation model, the performance of the semantic food name matching component, and the effectiveness of the personalized recommendation component. The obtained results provide an overall assessment of how well these components perform their intended tasks within the developed system.

4.3.1 Performance of the YOLO-Based Food Detection and Segmentation Model

The trained YOLO-based model was evaluated to assess its ability to detect and segment food items from meal images. The training and validation results were analyzed to observe the learning behavior of the model throughout the training process. Model performance was further evaluated using metrics such as precision, recall, mAP, and relevant segmentation metrics. These results provide an assessment of the model's ef-

fectiveness in accurately identifying food categories and generating their corresponding segmentation masks.

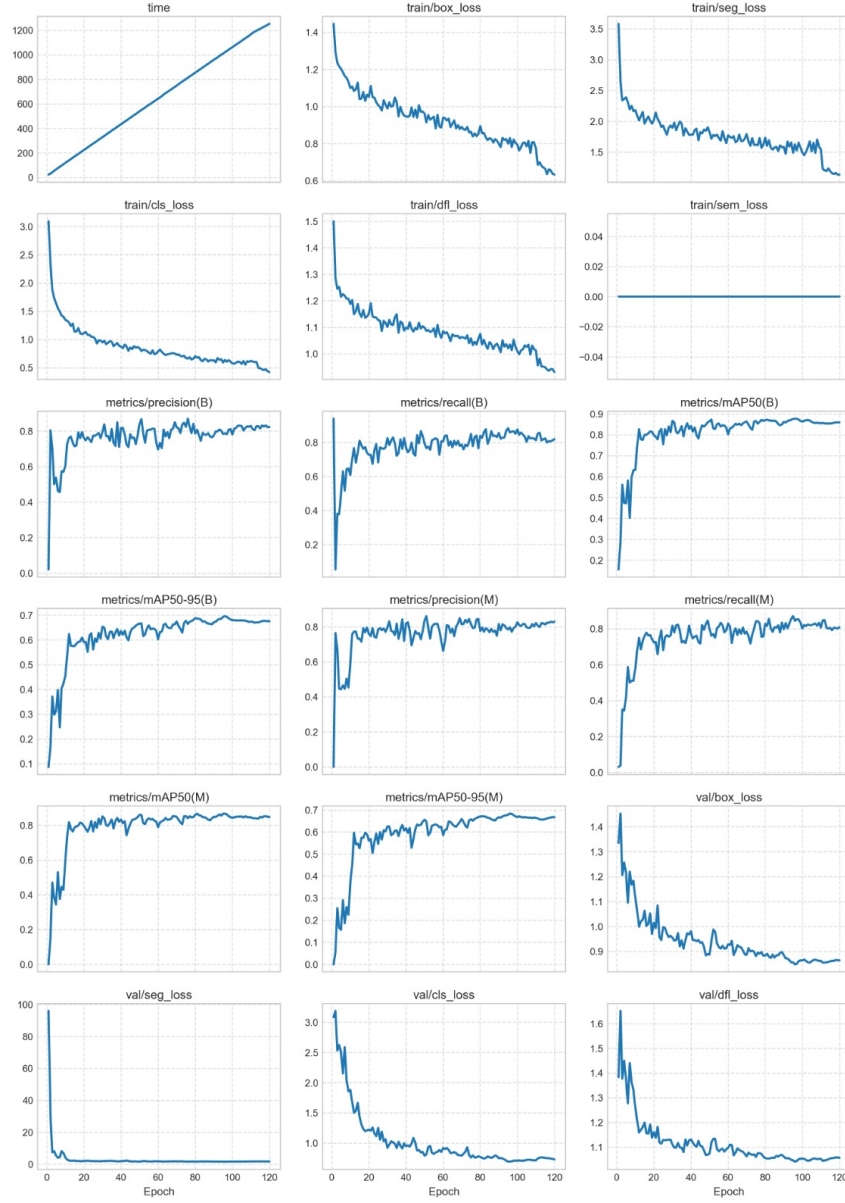


Figure 4.6: Training and Validation Performance of the YOLO-Based Food Detection and Segmentation Model

4.3.2 Semantic Food Name Matching Performance

The semantic food name matching component was evaluated to determine its accuracy in assigning the appropriate food name to each detected food item. The detected food labels were compared with the corresponding entries in the nutritional database, considering variations in local, common, and standardized food names.

Evaluation Measure	Value
Box Precision	0.8210
Box Recall	0.8344
Box mAP50	0.8578
Box mAP50-95	0.6736
Mask Precision	0.8331
Mask Recall	0.8026
Mask mAP50	0.8465
Mask mAP50-95	0.6562

Table 4.1: Performance Metrics of the YOLO-Based Food Detection and Segmentation Model

The evaluation was performed using a set of test cases, where the predicted food names were compared with their expected matches. The results were used to determine the accuracy of the matching process and to assess its ability to correctly associate detected food items with their relevant nutritional database entries.

```

===== TEST 1 =====
INPUT SENTENCE: 'maile 1 plate dal ra 1 plate bhat khaye'

IDENTIFIED FOOD ITEMS:
- [rice] Cooked Rice: 150.0g (195.0 kcal, 4.05g P, 42.3g C, 0.45g F)
- [dal] Cooked Lentils: 150.0g (174.0 kcal, 13.5g P, 30.15g C, 0.6g F)

TOTAL NUTRITION: {'calories': 369.0, 'protein': 17.55, 'carbs': 72.45, 'fat': 1.05}

===== TEST 2 =====
INPUT SENTENCE: 'aaja bihan maile 150 gram kukhura ko masu ani 200 gram chiura khayeko thiye'

IDENTIFIED FOOD ITEMS:
- [chicken] Cooked Chicken: 150.0g (235.5 kcal, 48.15g P, 0.0g C, 4.8g F)
- [chiura] Beaten Rice: 200.0g (700.0 kcal, 14.0g P, 152.0g C, 1.4g F)

TOTAL NUTRITION: {'calories': 935.5, 'protein': 62.15, 'carbs': 152.0, 'fat': 6.2}

===== TEST 3 =====
INPUT SENTENCE: 'khana ma alu ra palung saag thiyos'

IDENTIFIED FOOD ITEMS:
- [spinach] Spinach: 150.0g (34.5 kcal, 4.35g P, 5.4g C, 0.6g F)
- [potato] Boiled Potato: 150.0g (130.5 kcal, 2.85g P, 30.15g C, 0.15g F)

```

Figure 4.7: Semantic Food Name Matching

4.3.3 Personalized Recommendation Performance

The personalized recommendation component was evaluated to assess its ability to generate suitable diet and fitness recommendations according to the user's individual information. The generated recommendations were examined based on factors such as fitness goals, dietary preferences, health-related information, and the nutritional values obtained from the analyzed food items.

Different user scenarios were considered to determine whether the recommendations

changed appropriately according to the user's requirements. The evaluation indicates that the personalized dietary and fitness guidance provided by the recommendation component is based on the user's available profile, dietary preferences, fitness goals, and nutritional information.

4.4 Conclusion

The developed Nepali Lifestyle-Based Diet and Fitness Advisory System integrates food image analysis, nutritional analysis, and personalized recommendation into a unified system. The implemented system is capable of detecting and segmenting food items from meal images using the trained YOLO-based model and associating the detected items with their corresponding food names and nutritional information.

The retrieved nutritional data is then used to analyze the food composition and support the generation of personalized diet and fitness recommendations based on the user's profile, dietary preferences, fitness goals, and available health information. The generated results are presented through the system's health report, providing users with a consolidated view of their analyzed food and recommendations.

Overall, the developed system fulfills the major objectives of the project and demonstrates the potential of combining computer vision, semantic food matching, nutritional analysis, and personalized recommendation techniques to support Nepali lifestyle-based dietary and fitness guidance.

4.5 Limitations

Although the developed system integrates food detection, segmentation, nutritional analysis, and personalized diet and fitness recommendation, several limitations were identified during its development and evaluation. The performance of the YOLO-based food detection and segmentation model depends on the size and diversity of the training dataset. Variations in food appearance, presentation, lighting, camera angle, background, and overlapping food items may affect the model's ability to generalize to unseen images.

The nutritional analysis depends on the availability and accuracy of the nutritional

database. Some Nepali foods, regional variations, and differences in food preparation may not be fully represented, which can affect the retrieved nutritional values. Similarly, the semantic food name matching component may produce less accurate results when unfamiliar, ambiguous, or closely related food names are encountered.

The personalized recommendation component depends on the accuracy and completeness of the user's information and the available nutritional data. In addition, the system is intended to provide general dietary and fitness guidance and is not a substitute for professional medical or nutritional advice. These limitations provide scope for further improvement through dataset expansion, nutritional database enrichment, and more extensive system evaluation.

4.6 Future Enhancement

The developed system can be further enhanced to improve its accuracy, reliability, and practical applicability. The food dataset can be expanded to include a wider range of Nepali and regional food categories, along with greater variations in food appearance, preparation, portion size, lighting, camera angle, and presentation. The nutritional database can also be enriched with additional local foods, different preparation methods, and more comprehensive nutritional information to improve the accuracy and coverage of nutritional analysis.

The food name matching and personalized recommendation components can be further improved to deliver more accurate, context-aware results. Future development can focus on more precise food portion estimation, improved nutritional calculation, Nepali-language interaction, and long-term dietary and fitness tracking. Integration with wearable and activity-monitoring devices could provide additional user data to generate more personalized recommendations and support continuous health and fitness monitoring.

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